



Securing API-to-API Communication: Zero Trust, Granular Access, and AI-Driven Anomaly Detection

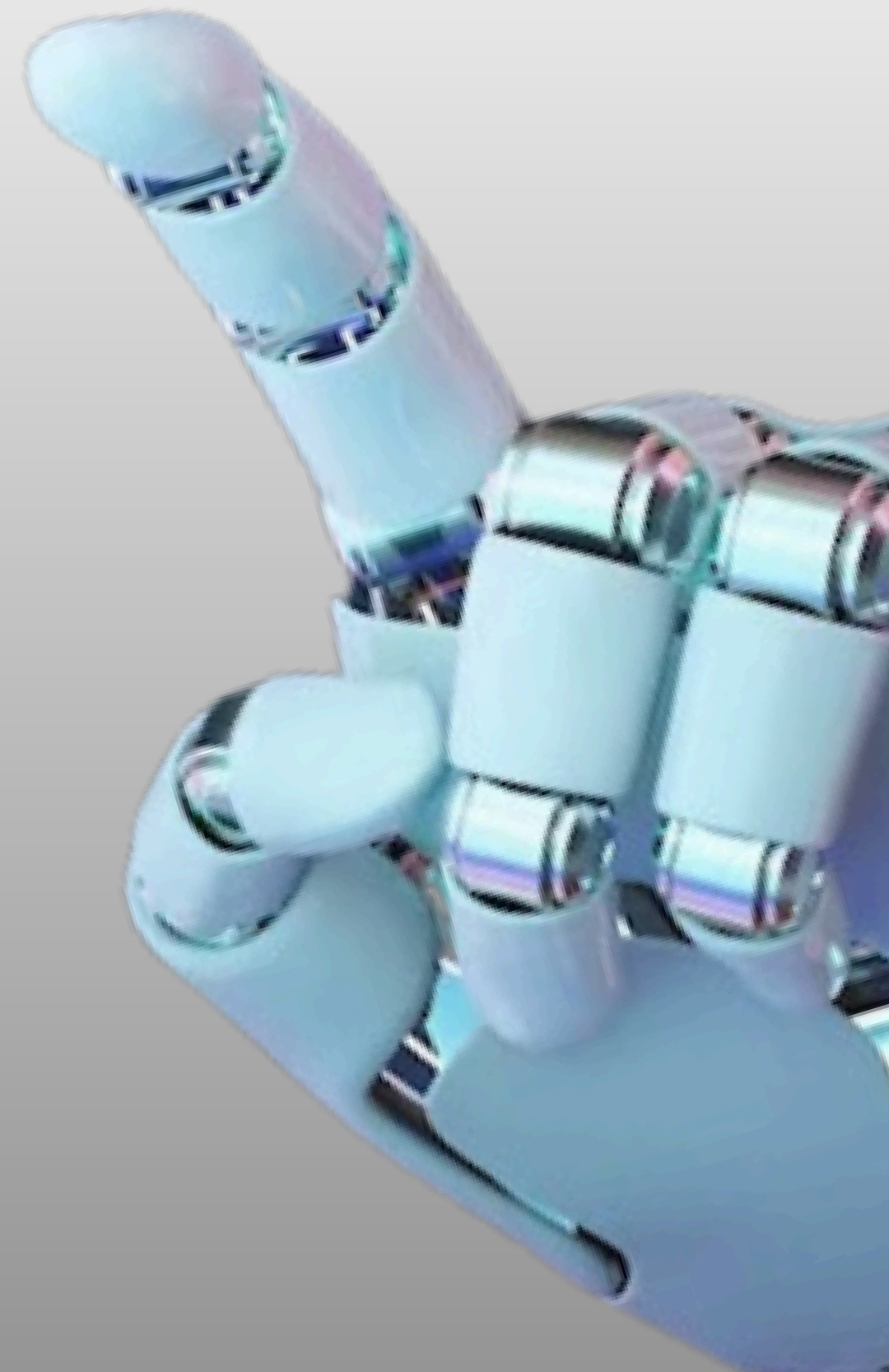
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OBJECTIVES



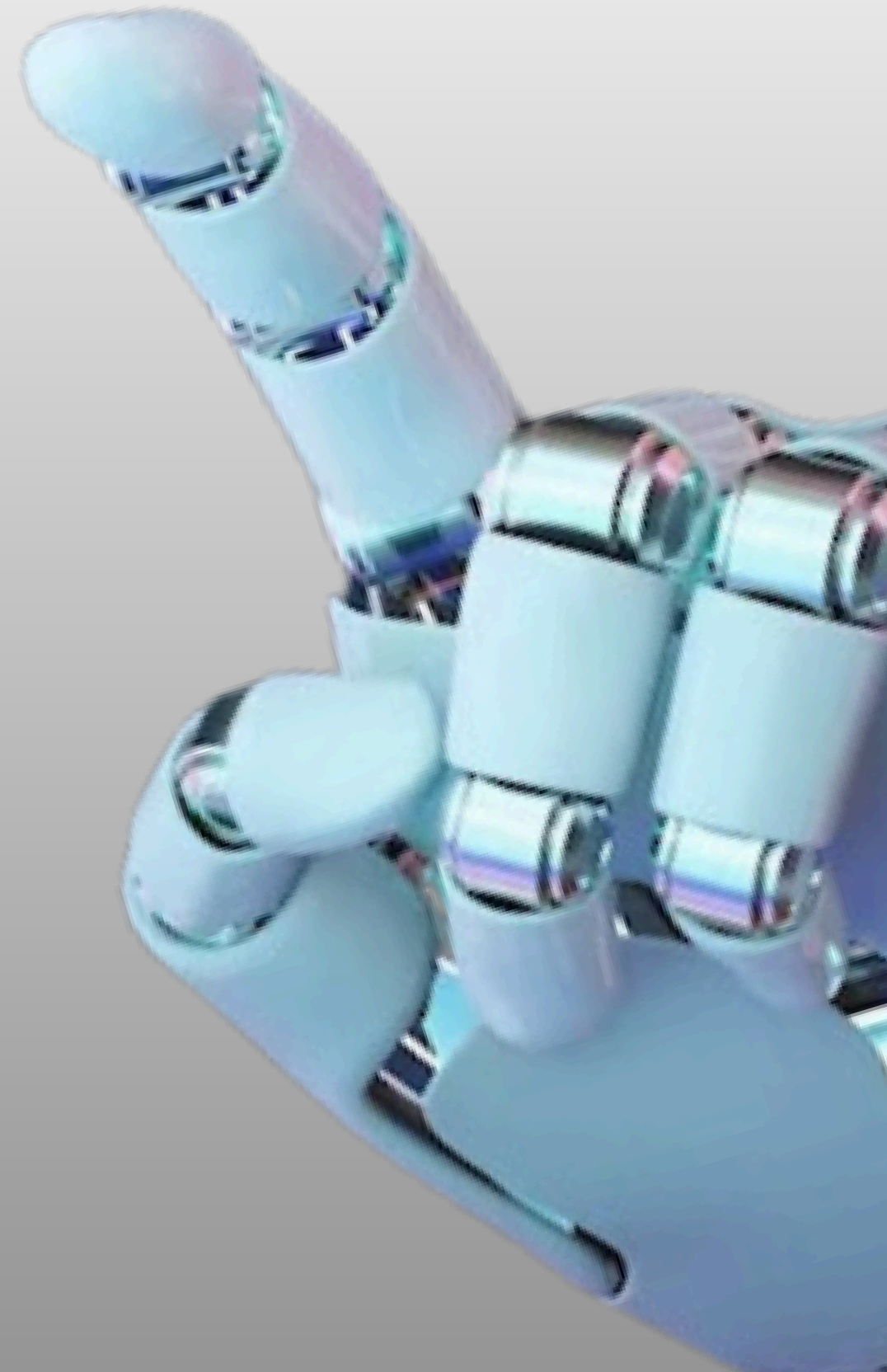
- 01 Implement Zero Trust Security Model for API-to-API communication
 - 02 Ensure secure access and data privacy in hospital management systems
 - 03 Utilize anomaly detection to identify potential security breaches
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OBJECTIVES



- 04 Enforce granular access control based on user roles and permissions
 - 05 Integrate Flask backend and Random Forest for real-time anomaly detection
 - 06 Protect sensitive healthcare data from unauthorized access and misuse
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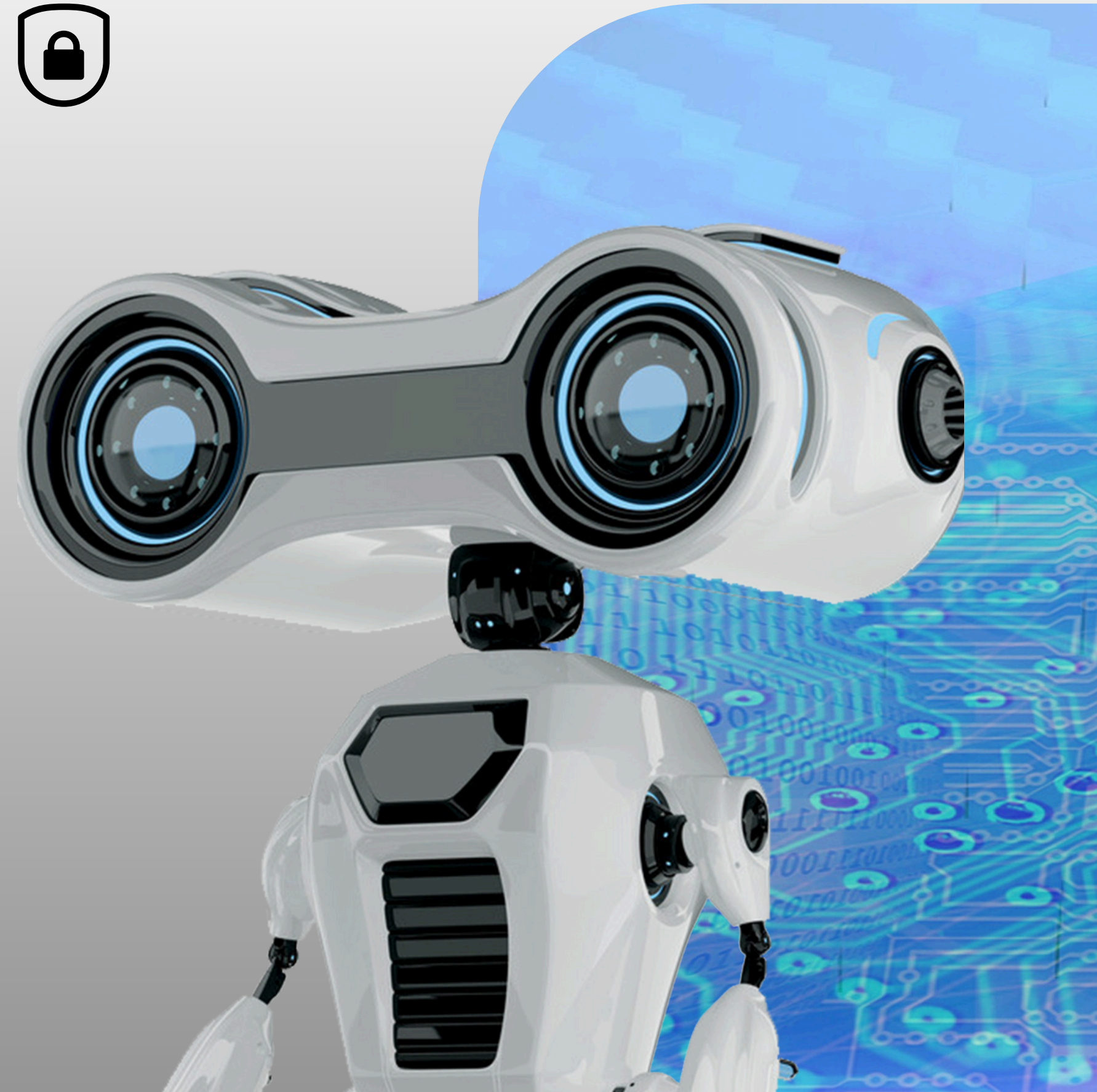
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Why AI Matters in Cyber Defense

Staying ahead in a dynamic threat landscape



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Detects threats
in real-time

Automates
responses

Identifies previously
unknown vulnerabilities

Where AI Meets Cyber Defense



Intelligent solutions for
complex problems

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AI's Edge in Cybersecurity

Strengthening defenses through innovation

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Processes vast
amounts of data
faster than
humans

Learns and
adapts to
emerging cyber
threats

Reduces false
positives,
ensuring
accurate alerts

Improves
incident response
times and
decision-making



AI vs Cybercriminals

A technological arms race

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Cybercriminals use AI to develop advanced malware and phishing attacks



AI-driven defenses counteract by analyzing and predicting malicious behaviors

Challenges in AI-Powered Cybersecurity

The double-edged sword of AI

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High costs and resource requirements for implementation

Risk of bias in training datasets leading to inaccurate detection

Potential misuse of AI by attackers



Responsible Use of AI

Balancing innovation with integrity

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A bank used AI to detect and prevent fraudulent transactions, saving millions



AI tools helped identify a ransomware attack, minimizing data loss for a corporation

Case Studies

Real-World Success Stories

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Integration of AI
with blockchain
for secure
identity
management

Development of
explainable AI to
build trust in
automated
decisions

Wider adoption of
AI-driven
automation in
small businesses

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The evolving
frontier

Future of AI in Cybersecurity





Defending the digital frontier with AI

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Thank You